

End-to-End Guide: Mock Anaplan Server & LightGBM Forecast Integration

This guide walks you through setting up and running your **Mock Anaplan Server**, pulling the exact Workers' Compensation input data from your image, running the **LightGBM ML forecasting engine** (`wc_lightgbm.py`), and pushing the **P10, P50 (Median), and P90** forecasts into Anaplan using **VS Code** and **Postman**.



Project File Structure

Place these files in a single folder (e.g., `anaplan_ml_project`):

```
anaplan_ml_project/
├── mock_anaplan_server.py      # Anaplan v2.0 REST API Mock Server
├── wc_lightgbm.py             # LightGBM Quantile Forecast Model
├── run_pipeline.py            # Automated Pull -> Train -> Push Pipeline
├── Anaplan_Mock_Postman_Collection.json # Postman Collection File
└── README.md                  # This manual
```

🔧 Step 1: Install Python Libraries (in VS Code)

1. Open **VS Code**.
2. Open your project folder (`File > Open Folder...`).
3. Open a terminal (`Terminal > New Terminal` or press `Ctrl + ``).
4. Run this command to install the required packages:

```
pip install fastapi uvicorn requests pandas numpy lightgbm
```

(Note: If you do not have *LightGBM* compiled, the script includes a built-in statistical quantile regressor that will run smoothly without errors).

🚀 Step 2: Start the Mock Anaplan Server

1. In your VS Code terminal, run:

```
python mock_anaplan_server.py
```

2. You will see:

```
=====
🚀 Starting Anaplan Mock Server on http://localhost:8000
Interactive Swagger UI Documentation: http://localhost:8000/docs
=====
```

3. **Leave this terminal running!** It acts as your local Anaplan Cloud server.
4. (Optional) You can open `http://localhost:8000/docs` in your browser to inspect all available endpoints in interactive Swagger UI.

Step 3: Run the Forecasting Pipeline

1. In VS Code, open a **Second Terminal** (Click the `+` button in the terminal pane).
2. Run the orchestrator script:

```
python run_pipeline.py
```

3. What happens automatically:

- **Step 1:** The script calls `/oauth2/token` and gets an OAuth token.
- **Step 2:** It downloads the 33 line-item features (Payroll, Claim Severity, Loss Development Factor, etc.) from Anaplan.
- **Step 3:** It triggers `wc_lightgbm.py` and calculates `P10`, `P50 (Median)`, and `P90` for **Jan-2026, Feb-2026, and Mar-2026**.
- **Step 4:** It converts predictions to CSV and uploads them via Anaplan's Chunk API (`/files/.../chunks/0`).
- **Step 5:** It triggers the Anaplan Import Process task and confirms completion.

Step 4: Testing & Interacting via Postman

If you prefer testing manually step-by-step with Postman:

1. Import the Collection

1. Open the **Postman** desktop app.
2. Click **Import** (top left).
3. Drag and drop `Anaplan_Mock_Postman_Collection.json` or browse to select it.
4. Click **Import**.

2. Execute Requests in Order:

#	Request Name	What it Does
1	1. Authentication - Get OAuth Token	Sends credentials, receives OAuth Bearer token, and automatically stores it in Postman variables.
2	2. Get Workspaces	Verifies your token against Anaplan workspace listings.
3	3. Get Models in Workspace	Retrieves the Workers' Comp pricing model context.
4	4. Get Input Feature View Data	Fetches the exact input features matching your Excel screenshot (Payroll, Claim Severity, Retention Rate, etc.).

- | | | |
|---|---|---|
| 5 | 5. Upload Forecasted Output Chunk | Pushes the ML predictions (P10, P50, P90) to Anaplan via the Chunk API. |
| 6 | 6. Run Import Process Task | Triggers the Anaplan process to commit the CSV into the model grid. |
| 7 | 7. Check Import Task Status | Confirms the task completed 100% with no errors. |
| 8 | 8. Inspect Anaplan Final Forecasted Output Grid | Returns the populated table in Anaplan to inspect the newly updated forecast. |

Mapping Reference (Snapshot vs API Data)

Input Data Extracted from Anaplan (Left Side of Image):

- **Dimensions:** Region (North East), Channel (Agency), Industry_Class (Agriculture), Account_Size (Small)
- **Time Periods:** 1/1/2022 , 2/1/2022 , 3/1/2022
- **Key Features:**
 - Payroll : 54,915,332.36 → 56,315,326.01 → 59,848,555.91
 - Employee_Count : 1,434 → 1,357 → 1,418
 - Historical_Loss_Ratio : 79.8% → 84.1% → 75.8%
 - Claim_Severity : \$31,605 → \$26,910 → \$27,747
 - GWP : \$856,055.57 → \$971,752.92 → \$1,120,470.33

Output Forecast Generated & Pushed to Anaplan (Right Side of Image):

- **Dimensions:** Region (Mid West), Channel (Agency), Industry Class (Agriculture), Account Size (Large)
- **Forecast Output:**
 - Jan-2026: P10: \$39,850 | P50 (Median): \$36,879 | P90: \$47,557
 - Feb-2026: P10: \$44,207 | P50 (Median): \$34,222 | P90: \$48,144
 - Mar-2026: P10: \$39,483 | P50 (Median): \$36,949 | P90: \$44,309